



BAULKHAM HILLS HIGH SCHOOL

Half -Yearly 2017
YEAR 11 ADVANCED TASK 1

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 1.5 hours
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-15
- Marks may be deducted for careless or badly arranged work

Total marks – 76

Exam consists of 9 pages.

This paper consists of TWO sections.

Section 1 – (10 marks) Pages(2-4) **Questions 1-10**

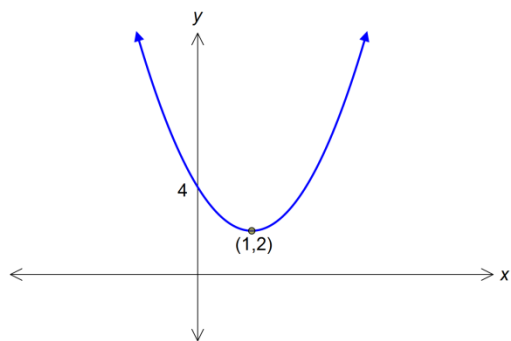
- Attempt Question 1-10
- Answer on answer sheet provided

Section II – (66 marks) Pages(5-9) **• Attempt questions 11-15**

Section I - 10 marks**Use the multiple choice answer sheet for question 1-10**

1. The factorisation of $x^3 - 8$ is
- (A) $(x - 2)(x^2 - 2x + 4)$
(B) $(x - 2)(x^2 + 2x + 4)$
(C) $(x - 2)(x^2 - x + 4)$
(D) $(x - 2)(x^2 + x + 4)$
2. The solutions to the equation $x^2 - 5x + 2 = 0$ are :
- (A) $\frac{5 \pm \sqrt{17}}{2}$
(B) $\frac{-5 \pm \sqrt{17}}{2}$
(C) $\frac{5 \pm \sqrt{33}}{2}$
(D) $\frac{-5 \pm \sqrt{33}}{2}$
3. Which of the following is equivalent to $\frac{1}{2\sqrt{5} - \sqrt{3}}$?
- (A) $\frac{2\sqrt{5} - \sqrt{3}}{7}$
(B) $\frac{2\sqrt{5} + \sqrt{3}}{7}$
(C) $\frac{2\sqrt{5} - \sqrt{3}}{17}$
(D) $\frac{2\sqrt{5} + \sqrt{3}}{17}$
4. $\frac{8^{n+1}}{2^{n-2}} =$
- (A) 4^{-1}
(B) 4^3
(C) 2^{2n+1}
(D) 2^{2n+5}

5.



The equation for the given parabola is:

(A) $y = (x + 1)^2 + 2$

(B) $y = (x - 1)^2 + 2$

(C) $y = (x - 2)^2 + 1$

(D) $y = 2(x - 1)^2 + 2$

6. How many solutions of the equation $(\sin x - 1)(\tan x + 2) = 0$ lie between 0° and 360° ?

(A) 1

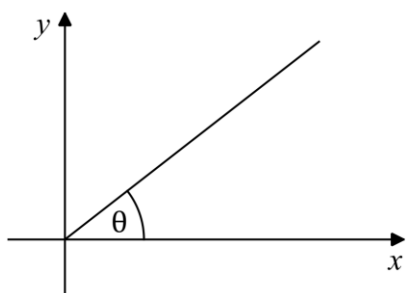
(B) 2

(C) 3

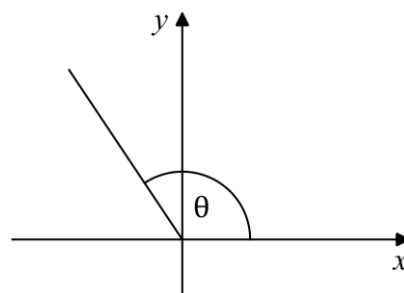
(D) 4

7. For the angle θ , $\sin \theta = \frac{7}{25}$ and $\cos \theta = -\frac{24}{25}$. Which diagram best shows the angle θ ?

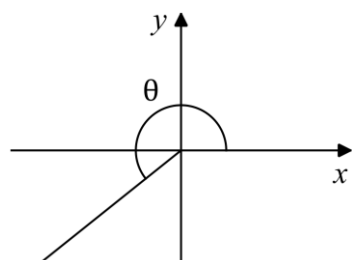
(A)



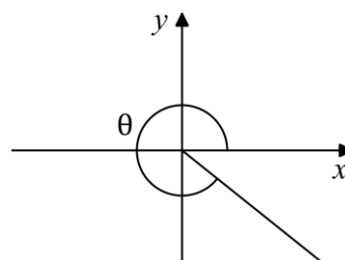
(B)



(C)

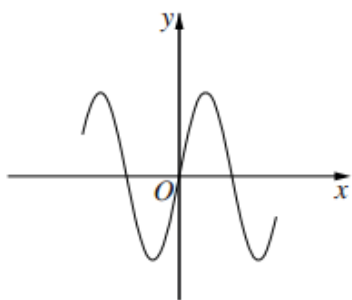


(D)

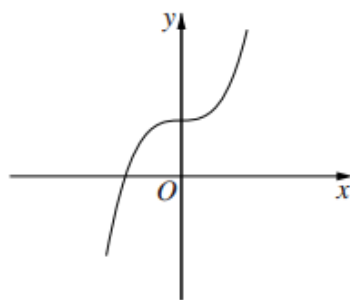


8. Which diagram shows the graph of an odd function?

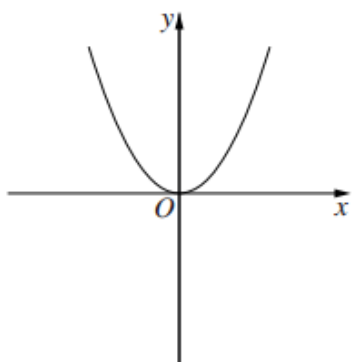
(A)



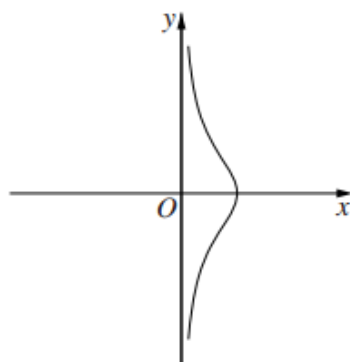
(B)



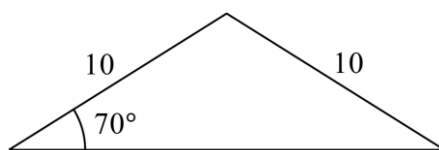
(C)



(D)



9.



NOT DRAWN TO SCALE

The area for the given triangle, correct to two decimal places, is:

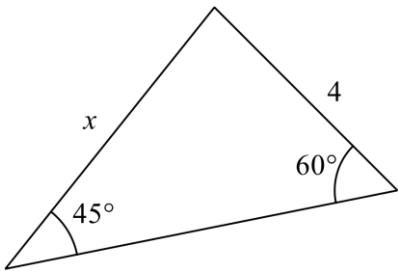
- (A) 38.30
- (B) 46.98
- (C) 32.14
- (D) 43.30

10. How many solutions does the equation $|\cos(2x)| = 1$ have for $0^\circ \leq x \leq 360^\circ$?

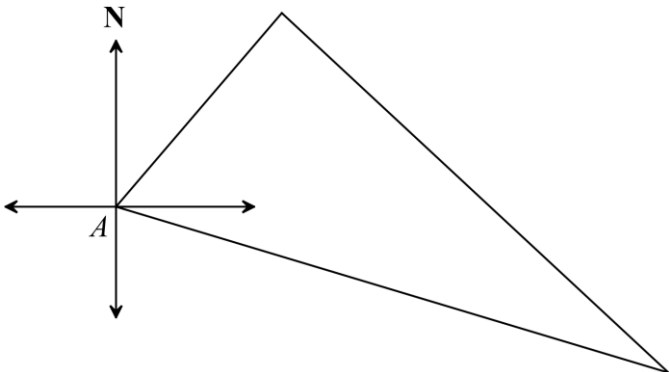
- (A) 1
- (B) 3
- (C) 4
- (D) 5

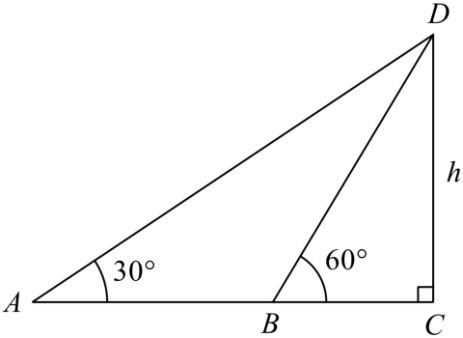
Section II – Extended Response**Attempt questions 11-15. All necessary working should be shown in every question.**

Question 11 (13 marks) Use the Question 11 section of the writing booklet.		Marks
a)	Solve	
	(i) $6 - \frac{2x+1}{4} = 3x$	2
	(ii) $ 3x - 1 = 6$	2
b)	Simplify	
	(i) $(x - 2)(x + 2) - (3 - x)$	2
	(ii) $\frac{1}{x} - \frac{1}{x-1}$	2
c)	If $(2\sqrt{3} - 2)^2 = a - \sqrt{b}$, find the values of a and b	2
d)	Council rates increased by 8% to \$1296. What were the rates prior to the increase?	1
e)	Solve simultaneously $4x + 6y = 11$ $17x - 5y = 1$	2
End of Question 11		

Question 12 (13 marks) Use the Question 12 section of the writing booklet.		
a)	If $f(x) = x^2 - 4x$, find: (i) $f(-2)$ (ii) $f(2\sqrt{3})$ (iii) the exact value of x, in simplest form, if $f(x) = 2$.	1 1 2
b)	Decide if the function $f(x) = (2x^2 - 5x)^2$ is even, odd or neither. Justify your answer	2
c)	Sketch the region in the Cartesian plane for which inequalities $y \leq 2x - 2$ $x^2 + y^2 \leq 4$ hold simultaneously	3
d)	Given the triangle below, find the exact value of x. 	2
e)	Find the exact value of $\tan \theta$ if $\cos \theta = \frac{\sqrt{2}}{5}$ and $270^\circ \leq \theta \leq 360^\circ$.	2
End of Question 12		

Question 13 (13 marks) Use the Question 13 section of the writing booklet.		Marks
a)	Consider the curve $y = \frac{2}{x-1} - 2$	
	(i) State the domain and range	2
	(ii) Find the intercepts	2
	(iii) Sketch the curve showing all important features	2
b)	Solve for $0^\circ \leq \theta \leq 360^\circ$	
	(i) $\sqrt{2} \sin \theta = 1$	2
	(ii) $2 \sin^2 \theta - \cos \theta = 1$	3
c)	Show that	2
	$\left(\frac{\cos \theta - \sin \theta}{\cos \theta} \right)^2 = \sec^2 \theta - 2 \tan \theta$	
End of Question 13		

Question 14 (13marks) Use the Question 14 section of the writing booklet.		
a)	Solve $ 2x + 5 = 3x + 9$	3
b)	Factorise $4x^3 - 12x^2 - x + 3$	2
c)	Given $f(x) = \begin{cases} x + 2 & \text{for } x \leq -2 \\ \sqrt{4 - x^2} & \text{for } -2 < x < 2 \\ 2 - x & \text{for } x \geq 2 \end{cases}$ (i) Sketch the function (ii) Hence or otherwise find the range of $f(x)$.	2 1
d)	A hiker left camp A and walked 15 km on a bearing of $N32^\circ E$ to B. He then turned and walked for 25 km to the point C, then 35 km back to A.  i) Redraw the diagram into your booklet showing the given information. ii) Find the size of $\angle ABC$. iii) Hence or otherwise find the bearing of B from C.	1 2 2
End of Question 14		

	Question 15 (14 marks) Use the Question 15 section of the writing booklet.	Marks
a)	Solve $\left(\frac{15}{x} + x\right)^2 - 11\left(\frac{15}{x} + x\right) + 24 = 0$	3
b)	Find an expression for the exact length of AB in terms of h . 	2
c)	i) Prove that $\tan A \sin A + \cos A = \sec A$	2
	ii) Hence or otherwise solve $\tan A \sin A + \cos A = \operatorname{cosec} A$ for $0^\circ \leq A \leq 360^\circ$	2
d)	If $f(x) = 2 - x^2$ and $g(x) = 2x - 1$	1 1 3
	i) Find $f(g(5))$.	
	ii) Show that $f(g(x)) = -4x^2 + 4x + 1$.	
	iii) Find the value(s) of x for which $f(g(x)) = g(f(x))$.	
	End of Exam	

① $x^3 - 8 = (x-2)(x^2 + 2x + 4)$ ⑧

② $x^2 - 5x + 2 = 0$
 $\therefore x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(2)}}{2(1)} = \frac{5 \pm \sqrt{17}}{2}$ ①

③ $\frac{1}{2\sqrt{5}-13} \times \frac{2\sqrt{5}+13}{2\sqrt{5}+13} = \frac{2\sqrt{5}+13}{20-17} = \frac{2\sqrt{5}+13}{3}$ ①

④ $\frac{8^{n+1}}{2^{n-2}} = \frac{(2^3)^{n+1}}{2^{n-2}} = \frac{2^{3n+3}}{2^{n-2}} = 2^{3n+3-n+2} = 2^{2n+5}$ ①

⑤ $y = a(x-m)^2 + n$ vertex $(1, 2)$
 $\therefore y = a(x-1)^2 + 2$ sub. $(0, 4) \therefore a = 2$ ①

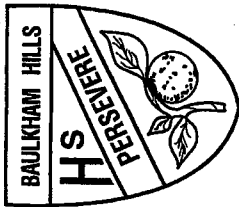
⑥ $\sin x - 1 = 0$ or $\tan x + 2 = 0$
 $\sin x = 1$ 1 solns.
 $\tan x = -2$ 2 solns. ②

⑦ $\sin \theta = \frac{7}{25}$ & $\cos \theta = -\frac{24}{25}$ $\therefore$
I or II second quadrant ②

⑧ Symmetrical around x & y-axis ①

⑨ $A = \frac{1}{2} ab \cdot \sin C$ $10 \times 40 \times \frac{10}{70}$
 $\therefore A = \frac{1}{2} \times 10 \times 10 \times \sin 40^\circ = 32.13938 \dots$ ②

⑩ $|\cos(2x)| = 1$ $0 \leq 2x \leq 360^\circ$
 $\pm \cos(2x) = 1$ $0 \leq 2x \leq 720^\circ$
 $\cos x = \pm 1$
 $x = 0^\circ, 180^\circ, 360^\circ, 540^\circ, 720^\circ \therefore 5 \text{ solns}$ ①



BAULKHAM HILLS HIGH SCHOOL
MATHEMATICS

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

Start Here →	1. A ☐	B ☒	C ☐	D ☐	6. A ☐	B ☐	C ☒	D ☐
	2. A ☒	B ☐	C ☐	D ☐	7. A ☐	B ☒	C ☐	D ☐
	3. A ☐	B ☐	C ☐	D ☒	8. A ☒	B ☐	C ☐	D ☐
	4. A ☐	B ☐	C ☐	D ☒	9. A ☐	B ☐	C ☒	D ☐
	5. A ☐	B ☐	C ☐	D ☒	10. A ☐	B ☐	C ☐	D ☒

Question 11 (13 marks)

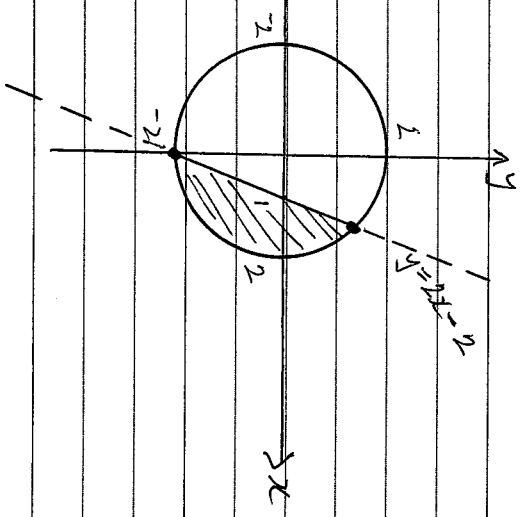
Marks

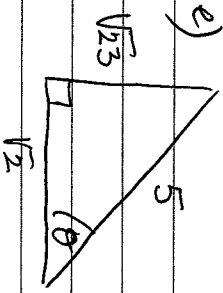
a) Solve (i) $6 - \frac{2x+1}{4} = 3x$	2 - correct solns.
$24 - (2x+1) = 12x$	1 - simplifying
$24 - 2x - 1 = 12x$	
$23 = 14x$	
$\frac{23}{14} = x$	
ii) $ 3x-1 = 6$	2 - correct solns
$3x-1 = \pm 6$	1 - one correct answer
$3x-1 = 6$	
$3x = 7$	
$x = \frac{7}{3}$	
$3x-1 = -6$	
$3x = -5$	
$x = -\frac{5}{3}$	
$\therefore$ solns. $x = \frac{7}{3}, -\frac{5}{3}$	
b) Simplify	
i) $(x-2)(x+2) - (3-x)$	2 - correct answer
$= x^2 - 4 - 3 + x$	1 - expanding one
$= x^2 + x - 7$	of brackets correctly

b) ii) $\frac{1}{x} - \frac{1}{x-1}$	Marks
$= \frac{x-1-x}{x(x-1)}$	2 - correct answer
$= \frac{-1}{x(x-1)}$	1 - correct denominator
or $\frac{-1}{x^2-x}$	
c) $(2\sqrt{3}-2)^2 = a - \sqrt{b}$	2 - correct values
$4 \times 3 - 8\sqrt{3} + 4 = a - \sqrt{b}$	for a and b
$16 - 8\sqrt{3} = a - \sqrt{b}$	1 - correct value
$16 - \sqrt{192} = a - \sqrt{b}$	for a or b
$\therefore a = 16, b = 192$	
d) 108% of $x = 1296$	1 - correct answer
$x = \$1200$	
$\therefore$ The rates were \$1200	

Question 12	Marks
a) $f(x) = x^2 - 4x$	2 - correct values for x & y
i) $f(-2) = (-2)^2 - 4(-2)$ $= 4 + 8 = 12$	1 - correct value for x or y
ii) $f(2\sqrt{3}) = (2\sqrt{3})^2 - 4(2\sqrt{3})$ $= 4 \times 3 - 8\sqrt{3}$ $= 12 - 8\sqrt{3}$ or $-1.8564...$	1 - correct elimination of x or y
iii) $f(x) = 2$ $x^2 - 4x = 2$ $x^2 - 4x - 2 = 0$ $x = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(-2)}}{2}$ $x = \frac{4 \pm \sqrt{24}}{2}$ $x = \frac{4 \pm 2\sqrt{6}}{2}$ $x = 2 \pm \sqrt{6}$	1 - making x or y the subject and subst. into the other equation 2 - correct solns in surd form simplified 1 - correct solns in surd form unsimplified

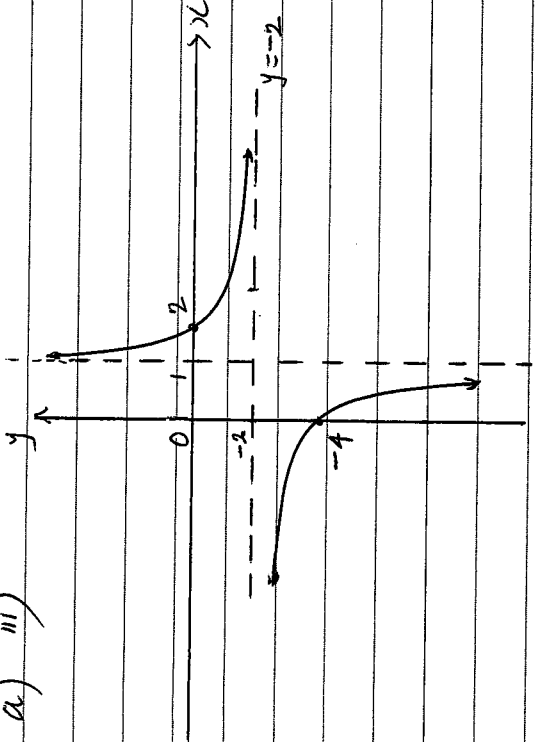
Question 12	Marks
a) $f(x) = x^2 - 4x$	1 - correct answer
i) $f(-2) = (-2)^2 - 4(-2)$ $= 4 + 8 = 12$	1 - correct answer
ii) $f(2\sqrt{3}) = (2\sqrt{3})^2 - 4(2\sqrt{3})$ $= 4 \times 3 - 8\sqrt{3}$ $= 12 - 8\sqrt{3}$ or $-1.8564...$	1 - correct answer exact or decimal
iii) $f(x) = 2$ $x^2 - 4x = 2$ $x^2 - 4x - 2 = 0$ $x = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(-2)}}{2}$ $x = \frac{4 \pm \sqrt{24}}{2}$ $x = \frac{4 \pm 2\sqrt{6}}{2}$ $x = 2 \pm \sqrt{6}$	2 - correct solns in surd form simplified 1 - correct solns in surd form unsimplified

	Marks
b) $f(x) = (2x^2 - 5x)^2$	2 - correct
$f(-x) = (2(-x)^2 - 5(-x))^2$	working
$= (2x^2 + 5x)^2 \neq f(x)$	1 - gets the expression for $f(-x)$
$\therefore$ not even	
also $(2x^2 + 5x)^2 \neq -f(x)$	
$\therefore$ neither	
c)	3 - correct region
	2 - one correct region and boundary 2 - two correct boundaries 1 - one of the boundaries correct

	Marks
d) $\frac{x}{\sin 60^\circ} = \frac{4}{\sin 45^\circ}$	2 - correct answer
$\frac{x}{\frac{\sqrt{3}}{2}} = \frac{4}{\frac{1}{\sqrt{2}}}$	1 - both exact values correct
$x = \frac{4 \times \frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}} = \frac{4\sqrt{3}}{2} \times \sqrt{2}$	1 - uses sine rule correctly
$x = 2\sqrt{6}$	
e)	2 - correct answer
	1 - correct ratio for tan theta
$\therefore \theta$ in 1st quadrant	
$\therefore \tan \theta < 0$	
$\tan \theta = -\frac{\sqrt{23}}{\sqrt{2}}$ or $\tan \theta = -\sqrt{\frac{23}{2}}$	1 - correct sign for tan theta
OR $\tan \theta = -\frac{\sqrt{46}}{2}$	

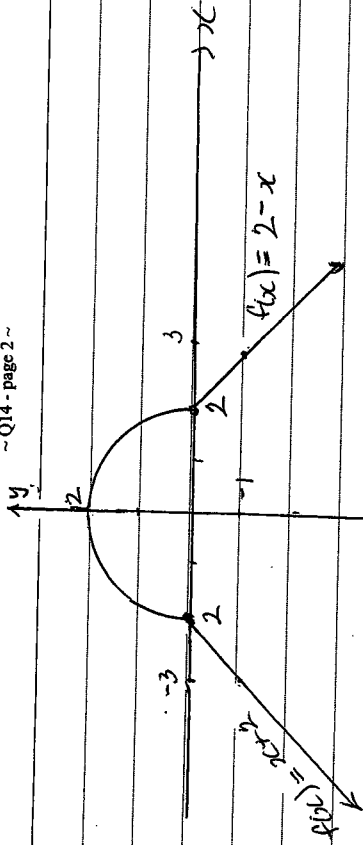
Question 13

a) $y = \frac{2}{x-1} - 2$	Marks
i) Domain = all real x , where $x \neq 1$ Range = all real y except $y = -2$	2 - correct solns. 1 - correct domain 1 - correct range
ii) x-intercept $\therefore y = 0$ $\therefore 0 = \frac{2}{x-1} - 2$ $2 = \frac{2}{x-1}$ $2(x-1) = 2$ $x-1 = 1$ $x = 2$	2 - correct x & y intercepts 1 - correct x or y-intercept
y-intercept $\therefore x = 0$ $\therefore y = \frac{2}{0-1} - 2$ $\therefore y = -4$	

a) iii)	Marks
	2 - correct shape with asymptotes and intercepts 1 - correct shape 1 - intercepts and asymptotes
b) i) $\sqrt{2} \sin \theta = 1$ $\sin \theta = \frac{1}{\sqrt{2}}$ $\therefore \theta = 45^\circ, 135^\circ$	2 - correct answer. 1 - correct acute angle
ii) $2 \sin^2 \theta - \cos \theta = 1$ $2(1 - \cos^2 \theta) - \cos \theta - 1 = 0$ $-2 \cos^2 \theta - \cos \theta + 1 = 0$ $2 \cos^2 \theta + \cos \theta - 1 = 0$ $(2 \cos \theta - 1)(\cos \theta + 1) = 0$ $\therefore \cos \theta = \frac{1}{2} \text{ or } \cos \theta = -1$ $\therefore \theta = 60^\circ, 300^\circ, 180^\circ$	3 - correct answer. 2 - uses correct identity & creates quad. trig. equation with one trig. ratio 2 - gets two sets of solns. from incorrect quad. eqn. 1 - uses correct trig. identity

Marks	
c) Show $\left(\frac{\cos \theta - \sin \theta}{\cos \theta} \right)^2 = \sec^2 \theta - 2 \tan \theta$	2 - correct proof
LHS = $\left(\frac{\cos \theta - \sin \theta}{\cos \theta} \right)^2$	1 - recognizes one of the trig identities and correctly expands bracket
$= \frac{(\cos \theta - \sin \theta)^2}{\cos^2 \theta}$	
$= \frac{\cos^2 \theta - 2 \cos \theta \sin \theta + \sin^2 \theta}{\cos^2 \theta}$	
$= \frac{\cos^2 \theta + \sin^2 \theta - 2 \cos \theta \sin \theta}{\cos^2 \theta}$	
$= \frac{1 - 2 \cos \theta \sin \theta}{\cos^2 \theta}$	
$= \frac{1}{\cos^2 \theta} - \frac{2 \cos \theta \sin \theta}{\cos^2 \theta}$	
$= \sec^2 \theta - \frac{2 \sin \theta}{\cos \theta}$	
$= \sec^2 \theta - 2 \tan \theta = \text{RHS}$	
$\therefore$ proven	

Question 14		Marks
a) Solve $ 2x + 5 = 3x + 9$	$\pm (2x + 5) = 3x + 9$	3 - correct solution
$2x + 5 = 3x + 9$	$-2x - 5 = 3x + 9$	
$-4 = x$ ①	$-14 = 5x$ ①	2 - finds both x-values correctly
① By checking $x \neq -4$ (not a solution)	$\therefore$ the only solution is $x = -\frac{14}{5}$	2 - finds one x-value correctly and checks
b) Factorise		1 - finds one of the x-values correctly
$4x^3 - 12x^2 - x + 3$		2 - correct factors
$= 4x^2(x-3) - (x-3)$		
$= (x-3)(4x^2 - 1)$		1 - gets 2 of 3 correct factors
$= (x-3)(2x-1)(2x+1)$		
c) $f(x) = x + 2$	$x \mid -3 \mid -2$	
	$2x \mid -1 \mid 0$	
$f(x) = \sqrt{4-x^2}$	semi circle	2 - all three sections correct
$f(x) = 2-x$	$x \mid 2 \mid 3$	
	$2-x \mid 0 \mid -1$	
		1 - two of the three sections of the piecewise function correct



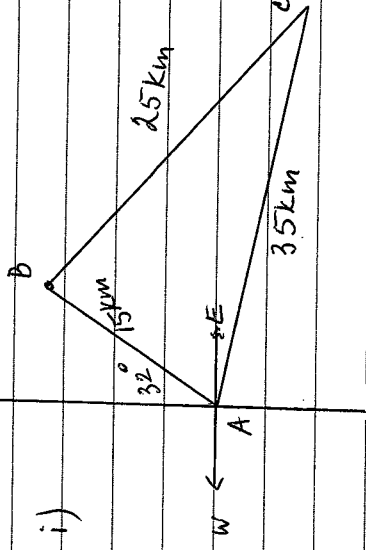
Marks

ii) Range - all real y where $y \leq 2$.

1 - correct answer

Marks

d) i)



1 - correct diagram

ii) by cosine rule

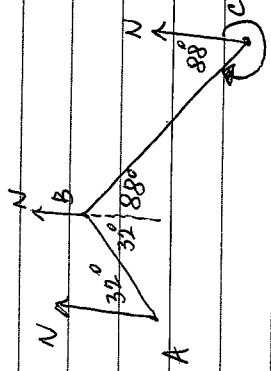
$$\cos(\angle ABC) = \frac{15^2 + 25^2 - 35^2}{2 \times 15 \times 25}$$

$$\cos(\angle ABC) = -\frac{1}{2}$$

$$\therefore \angle ABC = 120^\circ$$

2 - correct answer
1 - uses cosine rule correctly

iii)



2 - correct answer

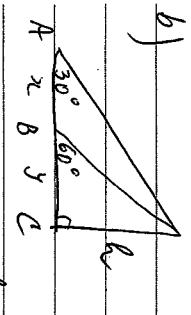
1 - recognises one correct pair of angles on parallel lines

bearing of B from C

$$360^\circ - 88^\circ = 272^\circ T$$

or $N 88^\circ W$

Question 15

	Marks
a) solve $\left(\frac{15}{x} + x\right)^2 - 11\left(\frac{15}{x} + x\right) + 24 = 0$	3 - correct answers
let $\frac{15}{x} + x = m$	2 - uses substitution to lead to quadratic equation and solves it correctly then significant progress towards solution
$\therefore m^2 - 11m + 24 = 0$	
$(m-8)(m-3) = 0$	
$m = 8$ or $m = 3$ ①	
$\therefore \frac{15}{x} + x = 8$	
$15 + x^2 = 8x$	
$\therefore x^2 - 8x + 15 = 0$	1 - uses substitution correctly
$(x-5)(x-3) = 0$	
$x = 5, x = 3$	
① $\therefore$ solutions $x = 5, x = 3$	
b)	
	2 - correct expression
let $AB = x$	
$BC = y$	1 - uses correct exact values of $\tan 30^\circ$ & $\tan 60^\circ$ to find $AB \times BC$
(1) $\tan 30^\circ = \frac{y}{x}$	
(2) $\tan 60^\circ = \frac{x}{y}$	
$\therefore \frac{1}{\sqrt{3}} = \frac{y}{x}$	
$\therefore y = \frac{x}{\sqrt{3}}$	
subst. $y = \frac{x}{\sqrt{3}}$ into (2)	
$\frac{1}{\sqrt{3}} = \frac{h}{x + \frac{h}{\sqrt{3}}}$	
$\frac{1}{\sqrt{3}} = \frac{h}{x + \frac{h}{\sqrt{3}}}$	

15b) cont.	
$x + \frac{h}{\sqrt{3}} = \sqrt{3}h$	
$\therefore x = \sqrt{3}h - \frac{h}{\sqrt{3}}$	where $x = AB$
OR $x = h\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)$ OR $h\left(\frac{3-1}{\sqrt{3}}\right)$	
OR $x = \frac{2}{\sqrt{3}}h$	
c) i) Prove $\tan A \sin A + \cos A = \sec A$	Marks
Proof: LHS = $\tan A \sin A + \cos A$	2 - correct proof
$= \frac{\sin A}{\cos A} \sin A + \cos A$	1 - significant progress towards solution
$= \frac{\sin^2 A}{\cos A} + \cos A$	
$= \frac{\sin^2 A + \cos^2 A}{\cos A}$	1 - uses correctly both identities $\tan A \times \sin A = \cos A$
$= \frac{1}{\cos A} = \sec A = \text{RHS}$	
$\therefore$ proven	

c) ii) Now Solve	
$\tan A \sin A + \cos A = \operatorname{cosec} A$	2 - correct solns.
for $0 \leq A \leq 360^\circ$	
from part (i) $\tan A \sin A + \cos A = \sec A$	1 - uses part (i) correctly
$\therefore$ Solve $\sec A = \operatorname{cosec} A$	1 - progress towards answers
$\frac{1}{\cos A} = \frac{1}{\sin A}$	1 - gets only one of the solns.
$\sin A = \cos A$	
$\therefore \tan A = 1$	
$\therefore A = 45^\circ, 225^\circ$	
d) $f(x) = 2 - x^2$, $g(x) = 2x - 1$	1 - correct answer
i) Find $f(g(5)) = f(2(5) - 1)$	
$= f(9) = 2 - 9^2 = -79$	
ii) $f(g(x)) = f(2x - 1)$	
$= 2 - (2x - 1)^2$	1 - correct solns.
$= 2 - (4x^2 - 4x + 1)$	
$= 2 - 4x^2 + 4x - 1$	
$= -4x^2 + 4x + 1$	

15d) cont.	
iii) Find $g(f(x)) = g(2 - x^2)$	3 - correct answer
$= 2(2 - x^2) - 1$	
$= 4 - 2x^2 - 1$	2 - finds $g(f(x))$ and equates
$\therefore g(f(x)) = 3 - 2x^2$ ①	$f(g(x)) \neq g(f(x))$ correctly
Now solve $f(g(x)) = g(f(x))$	1 - finds $g(f(x))$
$\therefore -4x^2 + 4x + 1 = 3 - 2x^2$ ①	
$\therefore 0 = 2x^2 - 4x + 2$	
$0 = x^2 - 2x + 1$	
$0 = (x - 1)^2$	
$\therefore (x - 1) = 0$ ①	

